

Lecture 22: Green's Theorem

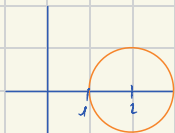
$$\oint_C \vec{F} \cdot d\vec{x}?$$

Green's Theorem: If C a closed curve, enclosing a region R , counterclockwise, $\vec{F}$ a vector field defined & differentiable in R , then $\oint_C \vec{F} \cdot d\vec{x} = \iint_R \text{curl } \vec{F} dA \Leftrightarrow \oint_C M dx + N dy = \iint_R (N_x - M_y) dA$

⚠ only for closed curves ⚠

Example: Let C = circle of radius 1 centered at $(2,0)$ counterclockwise

$$\oint_C y e^{x^2} dx + \left(\frac{1}{2} x^2 - e^{x^2} \right) dy \quad \text{D: rectly: } \begin{aligned} x &= 2 + \cos \theta & dx &= -\sin \theta d\theta \\ y &= \sin \theta & dy &= \cos \theta d\theta \end{aligned}$$



Ok, Green's Theorem:

$$\iint_R \text{curl } \vec{F} dA \Leftrightarrow \iint_R (N_x - M_y) dA = \iint_R \underbrace{(x + e^{-x^2})}_{N_x} - \underbrace{e^{x^2}}_{M_y} dA = \iint_R x dA \quad \bar{x} = \text{center of mass}$$

$$\iint_R x dA = \underbrace{\text{Area}(R)}_{\pi} \cdot \bar{x} = 2\pi \quad \bar{x} = \frac{1}{\text{Area}} \iint_R x dA = \left(\frac{1}{\text{Mass}} \iint_R x dA \right)$$

Special Case: If $\text{curl } \vec{F} = 0$ then $\vec{F}$ conservative? Green's Theorem: $\oint_C \vec{F} \cdot d\vec{x} = \iint_R \text{curl } \vec{F} dA = 0$ if $\text{curl } \vec{F} = 0$ This proves: if $\text{curl } \vec{F} = 0$ everywhere in R then $\oint_C \vec{F} \cdot d\vec{x} = 0$

Consequence: If $\vec{F}$ defined everywhere in the plane, and $\text{curl } \vec{F} = 0$, then $\vec{F}$ is conservative.

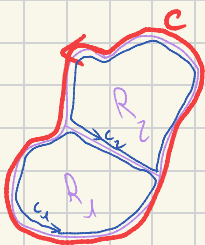
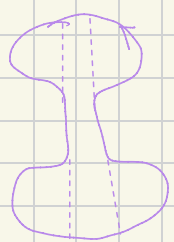
Proof: $\oint_C \vec{F} \cdot d\vec{x} = \iint_R \text{curl } \vec{F} dA = \iint_R 0 dA = 0$ Cannot apply Green's Theorem to vector field on Problem Set 8 Pblm 2 when C encloses the origin!

Proof of Green's Theorem: $\oint_C \vec{F} \cdot d\vec{x} = \iint_R (N_x - M_y) dA$ 1st Observation: we will prove: (Special Case with $N=0$) $\oint_C M dx = \iint_R -M_y dA$ A similar argument will show $\oint_C N dy = \iint_R M_x dA$

Summing, we get Green's Theorem.

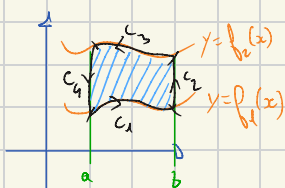
2nd Observation: We can decompose R into simpler regions

Cut R into "vertically simple" regions
 $a < x < b, f_1(x) < y < f_2(x)$



If we prove $\oint_{C_1} M dx = \iint_{R_1} -M_y dA$ and $\oint_{C_2} M dx = \iint_{R_2} -M_y dA$ then $\oint_C M dx = \oint_{C_1} + \oint_{C_2}$
 $\Leftrightarrow \iint_{R_1} + \iint_{R_2} = \iint_R -M_y dA$
because we go twice along the boundary between R_1 and R_2 with opposite directions

Main Step: prove $\oint_C M dx = \iint_R -M_y dA$



If R "vertically simple" and C = boundary of R counterclockwise

$$\oint_{C_1} M dx \quad y = f_1(x) \quad x \text{ from } a \text{ to } b \quad \Leftrightarrow \int_a^b M(x, f_1(x)) dx$$

$$\oint_{C_2} M(x, y) dx = 0; \text{ Same for } C_4, \oint_{C_4} M dx = 0 \quad x=b \quad dx=0$$

$$\oint_{C_3} M dx = \int_b^a M(x, f_2(x)) dx = - \int_a^b M(x, f_2(x)) dx$$

$$\oint_C M dx = \int_a^b M(x, f_1(x)) dx - \int_a^b M(x, f_2(x)) dx$$

$$\iint_R -M_y dA = - \int_a^b \int_{f_1(x)}^{f_2(x)} \frac{\partial M}{\partial y} dy dx$$

$$\text{Inner: } \int_{f_1(x)}^{f_2(x)} \frac{\partial M}{\partial y} dy = M(x, f_2(x)) - M(x, f_1(x)) \Leftrightarrow - \int_a^b M(x, f_2(x)) - M(x, f_1(x)) dx$$

"Imagine que le curl représente une densité de rotation de l'eau d'un lac, à quel point l'eau tourbillonne en chaque point. Green dit, si tu additionnes toute cette rotation sur toute la surface du lac tu obtiens exactement le "courant net" qui circule le long du bord du lac"